

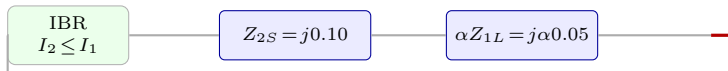
Positive sequence

relay CT: I_1, V_a



Δ winding:
blocks I_0

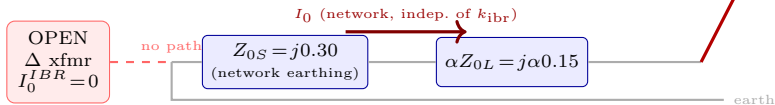
Negative sequence



Fault
 $V_a =$

SLG BC at fault:
 $V_a = 0, I_b = I_c = 0$
 $\Rightarrow I_1 = I_2 = I_0$
but $|I_1| \ll |I_0|$ for IBR

Zero sequence: I_0 from network earthing



$$k_0 = \frac{Z_{0L} - Z_{1L}}{3Z_{1L}} = \frac{0.15 - 0.05}{0.15} = 0.667 \quad (\text{line parameter, source-independent})$$

BUT: $|I_0| \approx 1.8 \text{ pu} \gg |I_1| = k_{ibr} \leq 0.20 \text{ pu} \Rightarrow$ relay over-estimates Z_m by 3–8×